SECTION B / 4 marks | 2 + 2

Q1. Scalar and vector products

FBISE Class 11 Physics Model Paper 2026 - complete descriptive solutions. All original and OR questions are included for preparation. In the examination, follow the paper's choice instructions.

Question. Using the commutative property, distinguish between the scalar product and vector product of two vectors. Give mathematical justifications.

Scalar product - commutative

The scalar product gives a scalar. If θ is the angle between two vectors,

$$\vec{A} \cdot \vec{B} = AB\cos\theta = BA\cos\theta = \vec{B} \cdot \vec{A}$$

Changing the order does not change the magnitude or sign. Therefore, the scalar product obeys the commutative law.

Vector product - anti-commutative

The vector product has magnitude AB sin θ and a direction perpendicular to both vectors, given by the right-hand rule. Reversing the order reverses this direction.

$$\vec{A} \times \vec{B} = AB\sin\theta\,\hat{n}, \qquad \vec{B} \times \vec{A} = -AB\sin\theta\,\hat{n}$$

$$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$$

Example. $\hat{i} \times \hat{j} = \hat{k}$, whereas $\hat{j} \times \hat{i} = -\hat{k}$. Thus, the vector product is anti-commutative. For parallel vectors, both cross products are zero.

OR - Inertial and non-inertial frames | 3 + 1 marks

A **frame of reference** is a coordinate system relative to which positions and motions are measured.

Property	Inertial frame	Non-inertial frame
Motion	At rest or moving with constant velocity relative to an inertial frame.	Accelerating or rotating relative to an inertial frame.
Newton's laws	Apply in their usual form without fictitious forces.	Require fictitious forces when written in their usual form.
Example	A train moving straight at constant velocity, approximately.	A braking train or rotating merry-go-round.

Reasoning. In a braking train, a passenger tends to maintain the original velocity because of inertia and moves forward relative to the train. The train is an accelerating, non-inertial frame.

SECTION B / 4 marks

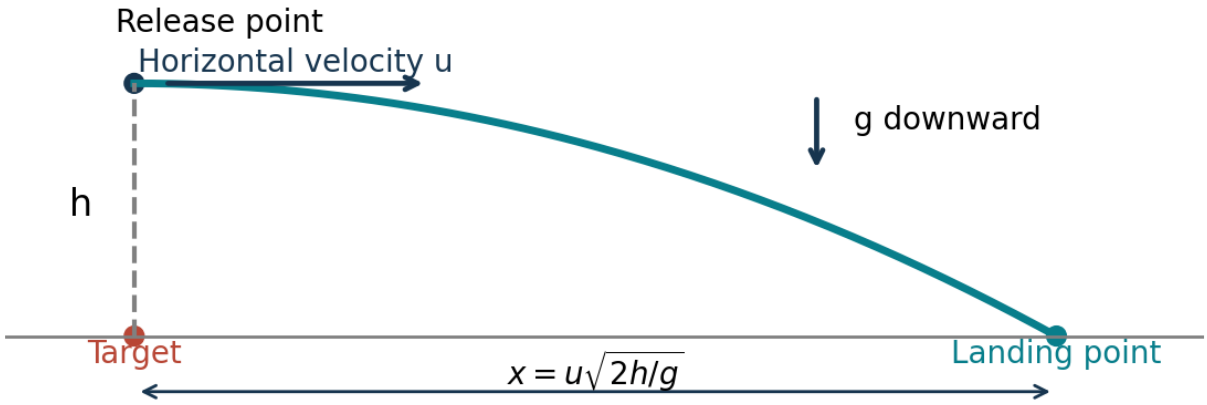
Q2. Bomb released from an aeroplane

Question. An aeroplane flying horizontally drops a bomb exactly above a target but misses it. Explain why.

Before release, the bomb has the same horizontal velocity **u** as the aeroplane. After release, gravity accelerates it downward. Neglecting air resistance, there is no horizontal force, so its horizontal velocity remains constant.

The bomb moves horizontally while falling vertically. It follows a **parabolic path** and lands ahead of the target.

$$t = \sqrt{\frac{2h}{g}}, \qquad x = ut = u\sqrt{\frac{2h}{g}}$$



Conclusion. For a stationary target, the bomb must be released a horizontal distance x before the aeroplane reaches the point directly above it. These equations assume level flight, zero initial vertical velocity and negligible air resistance.

OR - Checking a carbon-fibre concrete bridge | 4 marks

Concrete containing suitable electrically conducting carbon fibres can act as a **self-sensing material**. Its electrical resistance changes when it is strained or damaged.

1. Measure. Inspectors connect electrodes to the material and measure its electrical resistance using an appropriate instrument.

2. Compare. Readings under known loads are compared with the original, undamaged condition.

3. Interpret. Elastic strain can cause reversible resistance changes. Cracks or broken conducting paths can produce lasting changes, often an increase in resistance.

4. Assess. Abnormal changes help inspectors locate possible damage and decide where further inspection is needed. The measurement supports assessment without breaking the concrete open.

Technical reference: Chen and Chung, ACI Materials Journal 93(4), 1996, "Carbon fiber reinforced concrete as an intrinsically smart concrete for damage assessment during static and dynamic loading."

SECTION B / 4 marks per alternative

Q3. Ideal fluid and types of force

Question. Enlist the assumptions made about an ideal fluid.

Properties of an ideal fluid

1. Incompressible. Its density remains constant, even when pressure changes.

2. Non-viscous. It has no internal friction, so mechanical energy is not dissipated through viscosity.

Usual assumptions for elementary ideal-flow analysis

3. Steady flow. At a fixed point, the fluid velocity does not change with time.

4. Irrotational flow. Fluid elements have no local spinning motion about their own centres. This is a common additional assumption in the ideal-flow model.

Clarification. Incompressibility and zero viscosity describe the fluid itself. Steadiness and irrotationality describe its flow. Bernoulli's equation applies along a streamline for steady, incompressible, non-viscous flow with no added or removed mechanical energy.

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant along a streamline}$$

OR - Conservative and non-conservative forces | 2 + 2 marks

Feature	Conservative force	Non-conservative force
Definition	Work between two points is independent of the path taken.	Work depends on the path taken.
Closed path	Net work over a closed path is zero.	Net work over a closed path need not be zero.
Potential energy	A potential-energy function can be defined for the force.	Work cannot generally be described by a position-only potential energy.
Two examples	Gravitational force and electrostatic force.	Sliding friction and air resistance.

$$W_{\text{conservative}} = -\Delta U, \qquad \oint \vec{F} \cdot d\vec{r} = 0$$

Reasoning. Lifting an object between the same two heights gives the same work against gravity for every path. Sliding an object along a longer rough path generally produces a larger loss of mechanical energy due to friction.

SECTION B / 4 marks per alternative

Q4. Electrical power and magnetic force

Question. A device draws a current of 4 A when connected to a 12 V battery. Calculate its power and resistance.

Given V = 12 V and I = 4 A.

(a) Power consumed | 2 marks

Electrical power is the rate at which electrical energy is transferred.

$$P = VI = (12)(4) = 48 \text{ W}$$

The device consumes 48 joules of electrical energy per second.

(b) Resistance | 2 marks

Using Ohm's law, V = IR,

$$R = \frac{V}{I} = \frac{12}{4} = 3 \ \Omega$$

OR - Force on a current-carrying wire

Question. A wire of length 0.75 m carries 8 A in a uniform magnetic field of 0.2 T. It makes an angle of 45° with the field. Find the force, and then find the force when it is parallel to the field.

Given L = 0.75 m, I = 8 A, B = 0.2 T and θ = 45°.

(a) Wire at 45° to the field | 3 marks

$$F = BIL\sin\theta$$

$$F = (0.2)(8)(0.75)\sin 45^\circ = 0.8485 \text{ N}$$

$$F \approx 0.85 \text{ N}$$

The force is perpendicular to both the current direction and the magnetic field. Its direction is found using the appropriate hand rule.

(b) Wire parallel to the field | 1 mark

$$\theta = 0^\circ, \qquad F = BIL\sin 0^\circ = 0 \text{ N}$$

Reasoning. The force depends on the component of the wire perpendicular to the magnetic field. That component is zero when the wire is parallel to the field.

SECTION B / 4 marks per alternative

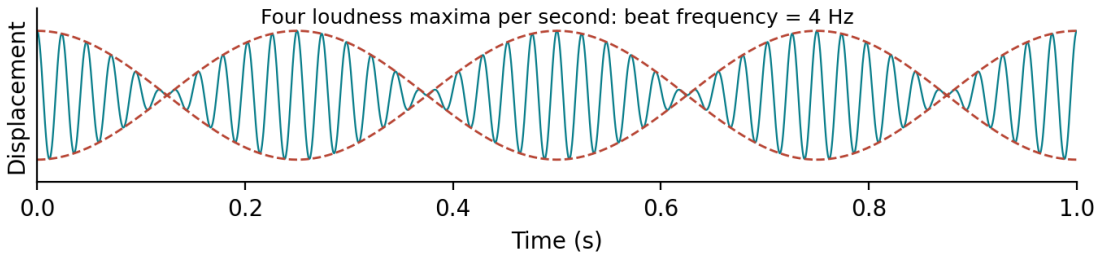
Q5. Beats and the refrigerator

Question. How are beats produced? Why are beats sometimes not heard distinctly?

Production. Beats are periodic increases and decreases in loudness caused by the superposition of two sound waves with slightly different frequencies and comparable amplitudes. The waves repeatedly become in phase and out of phase, producing constructive and destructive interference.

$$f_b = |f_1 - f_2|$$

For example, 256 Hz and 260 Hz produce 4 beats per second.

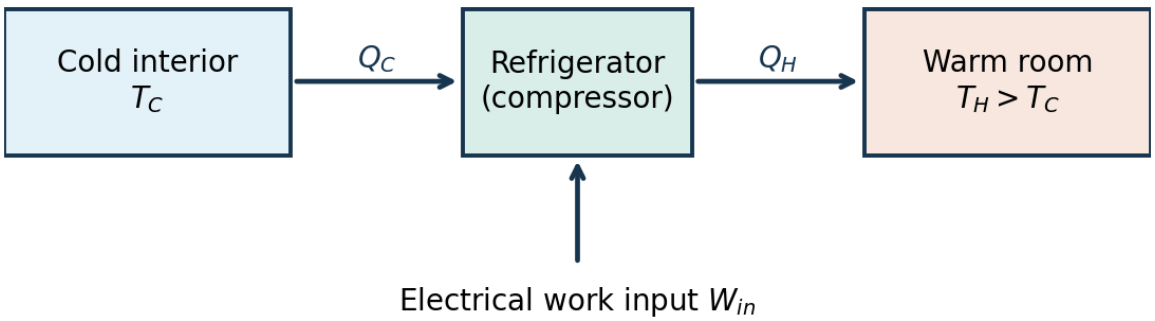


Why beats may be indistinct. If the frequency difference is too large, loudness changes occur too rapidly for the ear to distinguish separate beats. About 10 beats per second is a common textbook guide, rather than a sharp limit. Very unequal amplitudes also reduce the variation in loudness.

OR - Refrigerator as a heat engine in reverse | 3 + 1 marks

A heat engine receives heat from a hot reservoir, produces work and rejects heat to a cold reservoir. A refrigerator uses **external work** to transfer heat from its cold interior to the warmer surroundings.

Its evaporator absorbs Q_C from the cold interior. The electrically driven compressor supplies work W_{in}. The condenser rejects Q_H to the room.



$$Q_H = Q_C + W_{in}$$

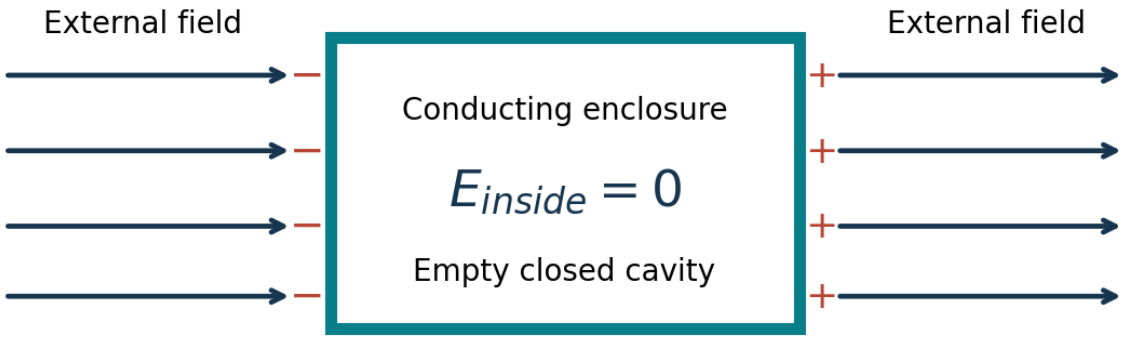
The rejected heat exceeds the heat removed from the interior because it also includes the work input. The external work enables heat transfer from the colder region to the hotter region.

SECTION B / 4 marks per alternative

Q6. Electrostatic shielding and interference

Question. A metal mesh enclosure protects a person inside from external electric charges. Name and explain the principle.

Principle - electrostatic shielding (Faraday-cage effect). A metal contains mobile electrons. An external electric field redistributes these charges over its surface. Their induced field opposes the applied field. At electrostatic equilibrium, an ideal closed conducting cavity containing no internal charge has zero electric field inside.



The drawing represents a closed conducting shell. A fine conducting mesh approximates this shielding. Openings allow some penetration, so its effectiveness depends on mesh geometry and distance from the openings.

OR - Path difference and bright/dark interference | 2 + 2 marks

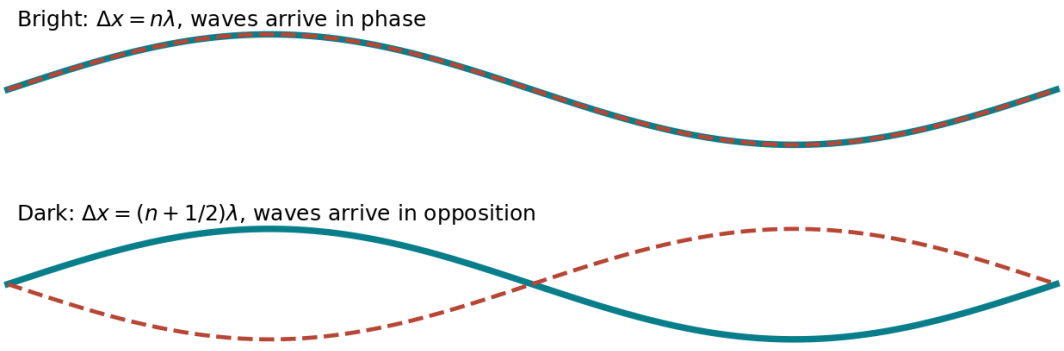
For two coherent waves of wavelength λ that start in phase, path difference Δx produces phase difference φ = 2πΔx/λ.

$$\text{Bright}: \quad \Delta x = n\lambda, \quad \phi = 2n\pi$$

Constructive interference. Crests meet crests and troughs meet troughs. The resultant amplitude and intensity are maximum.

$$\text{Dark}: \quad \Delta x = (n + \tfrac{1}{2})\lambda, \quad \phi = (2n + 1)\pi$$

Destructive interference. Crests meet troughs. The intensity is minimum. For equal amplitudes, ideal cancellation gives zero intensity. Here n = 0, 1, 2, ...



Phase comparison of the two waves at the observation point

SECTION B / 4 marks per alternative

Q7. Baryons, mesons and drag

Question. Define baryons and mesons. Give two examples of each.

Baryons | definition + examples

Baryons are hadrons consisting of **three valence quarks**. They participate in the strong interaction.

$$\text{Proton}: \ p = uud, \qquad \text{Neutron}: \ n = udd$$

The proton has charge +e and the neutron has zero net electric charge. Corresponding antibaryons contain three valence antiquarks.

Mesons | definition + examples

Mesons are hadrons consisting of a **valence quark-antiquark pair**. They also participate in the strong interaction.

$$\text{Positive pion}: \ \pi^+ = u\bar{d}, \qquad \text{Positive kaon}: \ K^+ = u\bar{s}$$

Particle	Constituents	Check of electric charge
Proton	u u d	(+2/3 + 2/3 − 1/3)e = +e
Neutron	u d d	(+2/3 − 1/3 − 1/3)e = 0
Positive pion	u and anti-d	(+2/3 + 1/3)e = +e
Positive kaon	u and anti-s	(+2/3 + 1/3)e = +e

OR - Drag force and three factors | 1 + 3 marks

Definition. Drag is the resistive force exerted by a fluid on an object moving relative to it. It opposes the object's motion relative to the fluid.

1. Relative speed. Drag generally increases when the speed relative to the fluid increases.

2. Size and shape. A larger frontal area usually increases drag. Streamlined shapes reduce the drag coefficient.

3. Fluid properties. Density and viscosity influence drag. The applicable relationship depends on the flow regime.

$$F_D = \frac{1}{2} C_D \rho A v^2$$

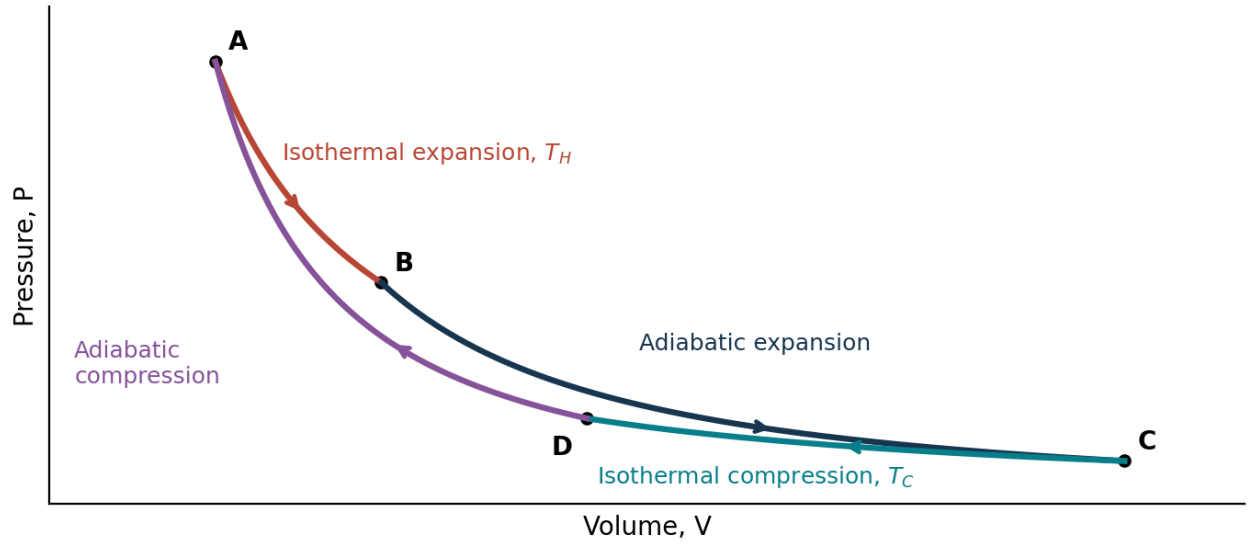
This common higher-Reynolds-number model shows the effects of drag coefficient C_D, density ρ, frontal area A and relative speed v. At sufficiently low speeds for a small sphere, viscous drag can instead be proportional to v.

SECTION C / 6 marks | diagram 2 + processes 4

Q1. Carnot cycle

Question. Draw a P-V diagram for a Carnot cycle and discuss its four processes.

A Carnot engine operates reversibly between a hot reservoir at T_H and a cold reservoir at T_C, with $T_H > T_C$. The working substance is taken as an ideal gas. Work W is positive when done **by** the gas.



1. A to B - Isothermal expansion at T_H

The gas absorbs heat Q_H from the hot reservoir and expands slowly at constant temperature. For an ideal gas, $\Delta U = 0$, so the heat absorbed equals the work done by the gas. Along this branch, $PV = nRT_H$ = constant.

2. B to C - Adiabatic expansion

The gas is insulated and continues expanding with $Q = 0$. It does work at the expense of its internal energy. Its temperature falls from T_H to T_C. The reversible adiabatic relation is PV^{γ} = constant, where $\gamma = C_P/C_V$.

3. C to D - Isothermal compression at T_C

The gas is compressed slowly while in contact with the cold reservoir. It rejects heat of magnitude Q_C at constant temperature. Its internal energy remains constant, and the magnitude of the work done on it equals Q_C.

4. D to A - Adiabatic compression

The gas is insulated and compressed with $Q = 0$. Work done on it increases its internal energy, raising its temperature from T_C to T_H. It returns to its initial state.

$$W_{net} = Q_H - Q_C, \qquad \eta = 1 - \frac{Q_C}{Q_H} = 1 - \frac{T_C}{T_H}$$

The clockwise enclosed area represents positive net work per cycle. Use kelvin in the efficiency equation. Since the gas returns to its initial state, its net change in internal energy over the cycle is zero.

SECTION C / 6 marks | derivation 4 + significance 2

Q1 OR. Quantized string frequencies

Question. Show that stationary-wave frequencies in a stretched string are quantized. Explain the physical significance.

Wave speed and fixed-end conditions

Consider a uniform string of length L, fixed at both ends, under tension F_T. Let μ be its mass per unit length. The transverse-wave speed is

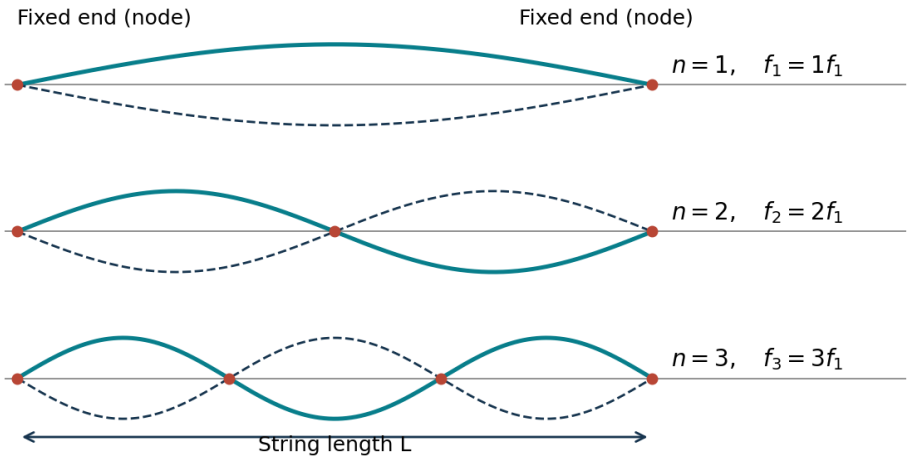
$$v = \sqrt{\frac{F_T}{\mu}}$$

Both fixed ends must be nodes. Adjacent nodes are separated by half a wavelength, so the string must contain an integer number of half-wavelengths.

$$L = n\frac{\lambda_n}{2} \quad \Longrightarrow \quad \lambda_n = \frac{2L}{n}, \qquad n = 1, 2, 3, \ldots$$

Using f = v/λ gives the permitted natural frequencies,

$$f_n = \frac{nv}{2L} = \frac{n}{2L}\sqrt{\frac{F_T}{\mu}} = nf_1$$



Red dots mark nodes. Solid and dashed curves show opposite extreme positions of the string, rather than two separate strings.

Mode	Wavelength	Frequency
Fundamental (n = 1)	λ₁ = 2L	f₁ = v/(2L)
Second harmonic (n = 2)	λ₂ = L	f₂ = 2f₁
Third harmonic (n = 3)	λ₃ = 2L/3	f₃ = 3f₁

Physical significance

The fixed-end conditions allow only the discrete natural frequencies f_1, $2f_1$, $3f_1$, ... An arbitrary wavelength cannot form a natural stationary mode with nodes at both ends. These harmonics explain the fundamental pitch and overtones of string instruments. This is quantization of the allowed classical modes.

SECTION C / 6 marks | field explanation 4 + permeability 2

Q2. Iron core inside a solenoid

Question. Why does inserting an iron core increase the magnetic field of a current-carrying solenoid? Explain the significance of relative permeability.

Magnetic field of an air-core solenoid

For a long solenoid, the approximately uniform magnetic field inside it is

$$B_0 = \mu_0 n I, \qquad n = \frac{N}{\ell}$$

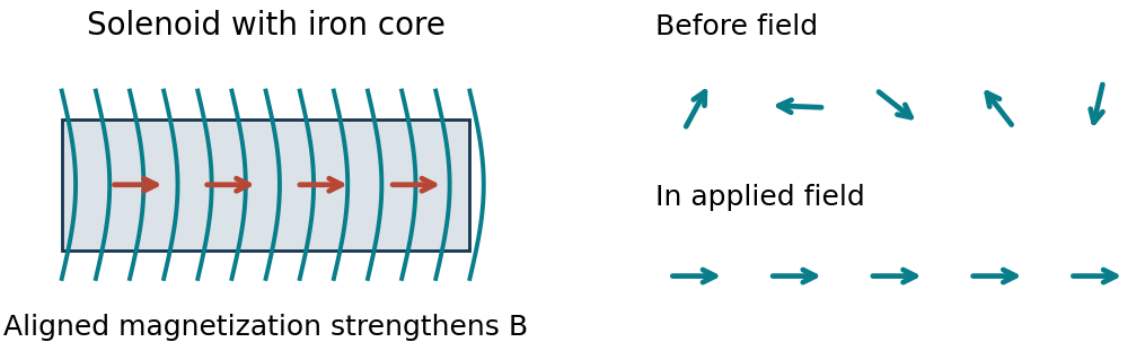
Here N is the number of turns, ℓ is the solenoid length, I is current, and μ_0 is the permeability of free space.

Why the iron core strengthens the field

1. Magnetic domains. Iron is ferromagnetic and contains regions called magnetic domains. In an initially unmagnetized piece, their magnetizations point in different directions, giving a small net magnetization.

2. Alignment. The solenoid's field favours domains aligned with the applied field. Those domains grow, and domain magnetizations turn towards the field.

3. Additional field. The core becomes magnetized. Its magnetization contributes a field in the same general direction as the solenoid's field, increasing the total flux density inside the solenoid.



Significance of relative permeability

Relative permeability compares a material's permeability μ with that of free space.

$$\mu_r = \frac{\mu}{\mu_0}$$

For an approximately linear material in a long solenoid,

$$B = \mu n I = \mu_0 \mu_r n I \quad \Longrightarrow \quad \frac{B}{B_0} \approx \mu_r$$

Iron has a large relative permeability under suitable conditions. Therefore, the same current can produce much more magnetic flux density with an iron core than with an air core.

The ratio is approximate for a real finite solenoid. Iron's permeability depends on field strength and magnetic history. Near saturation, its additional magnetization increases only slightly with the applied field.

SECTION C / 6 marks | definition 2 + statement 1 + explanation 3

Q2 OR. Moment of inertia and angular momentum

Question. Define moment of inertia. State and explain the law of conservation of angular momentum.

Moment of inertia | 2 marks

Moment of inertia measures a body's resistance to angular acceleration about a specified axis. It depends on the masses of its particles and their perpendicular distances r from the axis.

$$I = \sum_i m_i r_i^2, \qquad I = \int r^2\, dm \text{ for a continuous body}$$

Its SI unit is **kg m²**. Moving the same mass farther from the axis increases I because the distance is squared. For fixed-axis rotation, torque and angular acceleration are related by τ = Iα when I is constant.

Law of conservation | 1 mark

If the net external torque on a system is zero, its total angular momentum remains constant.

Mathematical explanation | 3 marks

$$\vec{\tau}_{ext} = \frac{d\vec{L}}{dt}$$

When the net external torque is zero, dL/dt = 0, so the total angular momentum does not change. For rotation about a fixed principal axis, L = Iω, giving

$$I_1\omega_1 = I_2\omega_2$$



Example. A spinning skater pulls her arms inward. Her moment of inertia decreases, so her angular speed increases. Extending her arms increases I and decreases ω. This assumes negligible external torque about the spin axis.

$$I_1 = 2\ \text{kg m}^2, \quad \omega_1 = 5\ \text{rad/s}, \quad I_2 = 5\ \text{kg m}^2$$

$$\omega_2 = \frac{I_1\omega_1}{I_2} = \frac{(2)(5)}{5} = 2\ \text{rad/s}$$

Angular momentum is conserved under the zero-external-torque condition. Rotational kinetic energy can change when internal work changes the mass distribution.